



OPEN ACCESS

Transforming healthcare: the PEACH Approach to reducing emissions and achieving net-zero

Casey Qian,¹ Fiona Parascandalo ,¹ Iliya Khakban ,¹ Sujane Kandasamy,² Russell de Souza,^{2,3,4} Myles Sergeant¹

¹McMaster University Faculty of Health Sciences, Hamilton, Ontario, Canada

²Department of Health Research Methods, Evidence, and Impact, McMaster University Faculty of Health Sciences, Hamilton, Ontario, Canada

³Mary Heersink School of Global Health and Social Medicine, McMaster University, Hamilton, ON, Canada

⁴Population Health Research Institute, Hamilton Health Sciences Corporation, Hamilton, ON, Canada

Correspondence to

Dr Myles Sergeant;
sergeam@mcmaster.ca

Received 10 February 2025
Accepted 9 August 2025

ABSTRACT

The healthcare sector has recognised its significant emissions and climate impact, and is beginning to address emission hotspots. However, implementing necessary changes while working with current stressors in the sector such as high patient volumes, limited resources, and staffing shortages, remains a challenge. PEACH Health Ontario (Partnerships for Environmental Action by Communities within Health care systems) was launched in 2021 to address this and has grown to a national scope of work with some of our initiatives. This paper outlines the 'PEACH Approach' to guide healthcare towards a net-zero future. This article describes how PEACH Health Ontario and the PEACH Approach were developed. We identify the various areas of healthcare sustainability that PEACH focuses on as well as our approach to collaboration and engagement across the sector. The PEACH Approach has led to the creation of specialty-specific green guidebooks, the Green Office Toolkit, and other knowledge mobilisation materials targeting system-wide transformation. These solutions are developed through multidisciplinary collaboration and knowledge translation, ensuring practical and evidence-based recommendations. The PEACH Approach drives a cultural shift in healthcare sustainability, creating solutions that lead to tangible outcomes. By using knowledge translation, providing practical solutions, and engaging with stakeholders, PEACH charts a course forward for both people and the planet.

INTRODUCTION

Over the past decade, the healthcare sector has recognised its emissions and climate impact and focused on identifying and beginning to address emission hotspots. Emission hotspots are processes responsible for the bulk of the upstream emissions, and where a change in technology or approach brings about the largest decrease in upstream emissions.¹ However, a major challenge remains: how can an overburdened industry and workforce effectively implement the transformations necessary to reduce environmental impact and continue to deliver high-quality patient care? Climate action in healthcare depends on innovative and practical solutions as well as internal champions and dedicated leadership.

PEACH Health Ontario (Partnerships for Environmental Action by Communities within Health Care systems; hereafter referred to as PEACH) launched in 2021 to address this difficult question and advance climate action in healthcare. Our approach and solutions are primarily focused on Ontario's healthcare system but have begun to

encompass some nationwide initiatives as well. In this article, we describe the 'PEACH Approach' to climate action and detail how it can lead healthcare towards a net-zero future. The PEACH Approach includes (1) working closely with experts from all sectors of the healthcare system to drive changes systemic change, (2) developing Green toolkits and guidebooks on various topics related to climate action in healthcare for healthcare providers and clinicians to adopt into their own practices, that are straightforward to adopt in their own practices, (3) ensuring that our work results in on-the-ground implementation, and (4) inviting input from new collaborators through promotion of our knowledge mobilisation materials.

History of PEACH

Through collaboration with the Canadian Coalition for Green Health Care (CCGHC)—a national organisation that aims to build capabilities and capacity in individuals and organisations to develop an environmentally sustainable, net-zero, and climate-resilient health system—a team of researchers, physicians and climate change experts identified a gap in how healthcare implemented mitigation, adaptation, and resiliency solutions in their facilities. This team consisted of Myles Sergeant P Eng, Linda Varangu, Sujane Kandasamy, and Zahira Khalid, FRCPC. The team worked together to identify key areas of a healthcare facility where innovation is needed to effectively engage in climate action. This was coined the 'ideal green facility' and encompassed eight core 'PEACH Pillars' central to reimagining a net-zero future for healthcare. These pillars include:

- Green leadership: Identifying and working with champions and leaders to create, implement, and reach net-zero goals as well as establish a culture of sustainability. Sustainability and climate leadership can be included in strategic plans and expanded on in site sustainability plans.
- Environmental education: As future healthcare workers are trained and current training models/materials are updated, it is important to include information on how to incorporate climate action into their practices.
- Sustainable supply chains: Determining areas of practice where supply chains could be shortened, products/materials could be eliminated or recycled, and where items could be safely reused with a focus on a circular economy approach.
- Sustainable drugs and devices: As drugs and devices are a major part of outpatient and



© Author(s) (or their employer(s)) 2025. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ Group.

To cite: Qian C, Parascandalo F, Khakban I, et al. *BMJ Leader* Published Online First: [please include Day Month Year]. doi:10.1136/leader-2025-001244

inpatient settings, this pillar focuses on reducing the impact of the top contributors of emissions —anaesthetic gases, overprescribing, metred-dose inhalers (MDIs) and pharmacy waste.

- ▶ Clean building design: Optimising building energy use through technology changes and practices. Also, strengthening facility climate resiliency with clean energy sources (eg, renewable energy—solar, wind) and building design that optimises energy and water usage.
- ▶ Environmentally conscious food procurement: Establishing sustainable menus with emission reduction considerations for food served to patients and staff, procurement and food waste reduction, and management.
- ▶ Green transportation: Fostering a culture of emissions reduction for personal travel and fleet management, including active transportation, efficient transportation, and virtual care options. Also, planning landscapes so they are safe for active transportation options and support electric vehicles.
- ▶ Integration of natural systems into healthcare infrastructure: Supporting adaptation and climate resilience by creating natural systems around healthcare buildings that contribute to local ecosystems and personal health and wellness.

PEACH strives to empower healthcare systems and their leadership bodies across Canada to adopt these pillars of sustainability, fostering a nationwide culture of environmentally responsible practice.

The ‘ideal green facility’ model led to the development of our PEACH Tree, a diagram comparing the impact of interventions on costs and greenhouse gas (GHG) emissions. It also outlines

the ‘low-hanging peaches’—action items that are important for efficient emission reductions in healthcare (Figure 1). During the development of the PEACH tree diagram, it was found that healthcare administrators may have an idea of what to do in order to reduce environmental impacts, but they did not know how to move forward with the necessary implementation actions.² With this, knowledge translation, leadership engagement, and practical solutions became a central focus for our team.

Recognising the need for PEACH to be uniquely impactful and not duplicate efforts of other organisations in the field, PEACH partners and collaborates with system leaders at the local, provincial, and national level. In 2024, PEACH formalised a partnership with McMaster University by establishing the PEACH office within the Department of Family Medicine (DFM). This partnership with McMaster University has allowed our team to expand our research and networking capabilities.

The Planetary Health collaboration within McMaster has supported PEACH’s development through funding of staff and conference attendance. PEACH now has a team of full-time and part-time research staff funded by both the DFM and the Planetary Health team. We continuously share ideas and collaborate on new projects with the faculty and staff involved with Planetary Health. Additionally, our team works closely with McMaster University’s Mary Heersink School of Global Health and Social Medicine. This partnership has allowed PEACH to expand its academic involvement with connections to many talented and passionate student volunteers who support research projects and publications. Our knowledge translation materials are strengthened by the Heersink School’s innovative researchers specialising

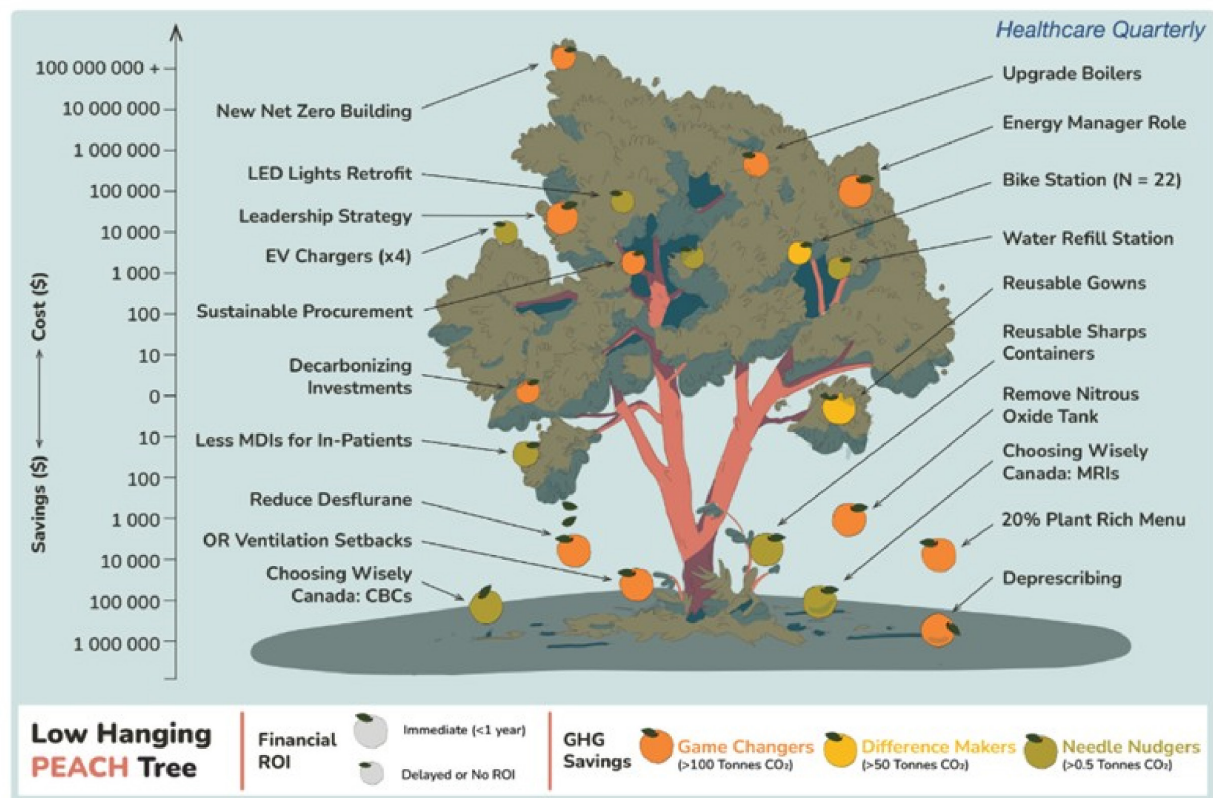


Figure 1 The PEACH tree diagram. Overview: The PEACH tree diagram comparing the impact of interventions on costs and GHG emissions across seven different categories. It depicts the ‘low-hanging fruit’ or quick, effective and low-cost actions that healthcare leadership can take in reducing emissions (eg, reduce desflurane) as well as other peaches on the tree that are expensive but impactful (eg, net-zero building), and some that are expensive and only ‘needle nudgers’ (eg, electric vehicle chargers). LED, light-emitting diode; GHG, greenhouse gas; CBC, complete blood count; ROI, return on investment; PEACH, Partnerships for Environmental Action by Communities within Health Care.

in social medicine, programme evaluation, and health inequalities. PEACH reaches out to not-for-profit organisations, charities, universities, and other organisations across Canada doing work in the area of sustainable healthcare to co-design materials and expand the reach of our initiatives. This can look like co-chairing a provincial or national committee, co-authoring a publication, hosting a panel discussion at a conference or webinar, holding monthly informal check-in calls, or sharing their work on our online platforms. It is important that productive and lasting partnerships are formed; thus, we are committed to cooperative collaboration and idea sharing.

In our capacity-building activities, PEACH also brought together environmental experts, medical practitioners, facility staff, community members, healthcare leadership, medical residents and students to establish a committed volunteer core. This engagement with volunteers is a continuous effort; we currently have close to 100 volunteers who support our many avenues of operations. Volunteers make up half of our executive team and 100% of our advisors. They are involved in all aspects of PEACH from conceptualisation to commercialisation, for example, students, medical residents, researchers and experts help the PEACH team plan, write and mobilise guidebooks and publications; volunteers plan and execute tree plantings; and volunteers assist in planning and facilitating large meetings and conferences. Their knowledge and involvement are central to our commitment to de-silo climate action within healthcare to develop a community of changemakers and climate champions.

Sustainability in healthcare requires multidisciplinary input, often leading to competition between disciplines with competing priorities. PEACH aims to be a positive example in the sector, de-siloing the field by bringing champions together to facilitate lasting change and spur innovation. The work of PEACH increases opportunities for collaboration across sectors by engaging healthcare workers and leaders in developing and sharing solutions.

PEACH Approach Overview

A cornerstone of the 'PEACH Approach' is the specialty-specific green guidebooks and toolkits created in collaboration with clinicians, researchers and leaders in sustainable healthcare. Each guidebook and toolkit targets a particular field or area of operation within healthcare and provides the reader with climate action recommendations so that environmental sustainability is considered alongside patient care in clinical decision-making and leadership engagement. The clinical recommendations are organised such that they correspond to one of the eight aforementioned PEACH Pillars. Authors and collaborators can personalise their guidebook by including additional information to better meet the unique demands of the medical specialty. For example, our Psychiatry Guidebook highlighted the negative implications of climate change for mental health and listed steps to mitigate climate anxiety.³ Some of our recent toolkits and guidebooks are part of a national project with the CCGHC, sponsored by Environment and Climate Change Canada. These materials are translated into French to help expand our audience to French-speaking Canada. Additionally, some of our toolkits and guidebooks (<https://peach.healthsci.mcmaster.ca/our-impact/guidebooks/>) have been disseminated through Choosing Wisely Canada (<https://choosingwiselycanada.org/climate/>).

To address the need for resources that contain practical steps for sustainable outpatient care, PEACH supported the development of the Green Office Toolkit which has undergone many revisions for its most recent edition (2023). The purpose of the

Green Office Toolkit is to provide guidance for clinicians who work in the outpatient setting to transform their clinical practice as well as advice for leadership on how to align their organisations with sustainability. The Green Office Toolkit highlights various domains of change for the reader to consider when transitioning their clinical practice, ranging from green prescribing, reuse of personal protective equipment, energy efficiency in the clinic and more.⁴

To highlight the practicality of the Green Office Toolkit, PEACH worked with the Hamilton Family Health Team's Green Initiative (HFHT-GI) to create the Green Office Challenge, which launched in the spring of 2024. Since 2019, HFHT-GI has been supporting environmental sustainability in the delivery of primary care both within clinical care and the waste and emission from the places in which care is provided. Their team summarises evidence, launches initiatives, delivers education events and training, and creates practical tools. The Green Office Challenge is a brief questionnaire that clinicians can use to grade the sustainability of their medical practice in eleven distinct areas as well as identify areas where future improvements can be made. The purpose of the Challenge is to engage with practitioners in hopes that new 'climate champions' can lead greater change in their communities nationwide.

The Green Office Toolkit focuses on effecting change at the level of clinicians and individual medical practices or departments. Other PEACH materials strive for hospital or health system-wide transformation, targeting hospital leadership and executives who can make these changes happen. These solutions are developed through interdisciplinary collaboration across various areas of hospital operations such as facility management, acute care, intensive care units (ICUs), emergency rooms, food services, corporate services, etc.

PEACH acknowledges that interdisciplinary and multidisciplinary input is tantamount to climate action, and this is at the forefront of our approach to any project. For instance, under the co-authorship of our Sustainable Food Committee, we recently released the Sustainable Health Care Food System Business Case and Implementation Guide, a document outlining the need for healthcare facilities to prioritise the transformation of food services into more sustainable models.⁵ The voices of healthcare professionals and academics are often the most prominent in conversations at the interface of food, climate, and health. However, PEACH drew on experts in food procurement and the institutional food service industry, sales and marketing in agriculture, and financial advisors in hospital supply chains to spearhead the Business Case and Implementation Guide. This ensured that our work included evidence-based, practical solutions that would appeal not only to C-suite level hospital executives to invest in, but also to facility workers who would implement them, while assuring healthcare professionals that food system overhauls are also best for their patients.

Knowledge translation is a central tool that PEACH uses to bring our solutions from research and articles to boardrooms and desks of stakeholders. Our Low-Hanging PEACH Tree illustrates the impact of various health facility interventions to reduce GHG emissions and their financial impacts to highlight 'low-hanging fruits', actions facilities can readily implement to reduce their carbon output, and in many cases, reduce spending.² By turning the abstract concept of reducing carbon output into quantifiable and readily implementable solutions, the Low-Hanging PEACH Tree makes sustainability more approachable to health facilities and demonstrates to decision-makers that sustainable interventions can mutually benefit patients and the planet.

For all PEACH research projects and products, after meaningful consultations with intersectoral experts and knowledge translation specialists, the final step involves the dissemination of the finished product. Our materials, no matter how robust their contents, mean little if they do not reach their desired audiences. To this end, PEACH has developed numerous streams of communication to expand our network not only to share findings with stakeholders, but also to meet and invite new collaborators. These streams include newsletters, social media marketing or word-of-mouth, publications with academic journals and news outlets, collaborative committee facilitation, webinars, and conference attendance. Through these collaborations, many of our products are living documents that continuously evolve because they are informed by state-of-the-art research that ensures sustainability in the healthcare field continues to flourish.

PEACH is currently conducting an evaluation of our guidebooks and toolkits to learn more about how they are being used and their impact. We are interviewing healthcare workers who have used our materials to understand how they are implemented and how we can more effectively reach more users. This project is scheduled to end in late 2025, and the results will inform how we create and share our future work. In the meantime, PEACH measures its success by establishing and tracking key performance indicators such as number of conferences, webinars and grand rounds presented at, media mentions, new partnerships formed and much more. By tracking these metrics, PEACH is able to review its annual impact and reflect on where more work is needed.

Healthcare workers who want to make changes in their field or place of work routinely contact our team to co-create materials, host events, or partner on publications. Due to our emphasis on approachability and de-siloing the field, practitioners feel comfortable reaching out to be part of our work or learn from our materials in all stages of their journey learning about healthcare and climate change. Every collaborator on our materials leaves the process knowing practical steps to lead change. They have the tools and knowledge needed to be a climate champion creating ripple effects in their field, facility, or across the healthcare sector.

CONCLUSIONS

The PEACH Approach drives a cultural shift in the work of healthcare sustainability and more broadly, the field of climate action, honouring input from all involved in making healthcare possible to create solutions that lead to tangible outcomes. Sustainability in healthcare can be an abstract and unwieldy concept for many, but the PEACH Approach of using knowledge translation, providing practical solutions, and engagement with

leadership and stakeholders demystifies the field and charts a course forward for people and planet.

Acknowledgements The authors would like to thank our many supporters within McMaster University who have provided funding and support for our staff as well as their expert guidance as we have developed, this includes Dr Heather Sheardown, Dr Paul OByrne, Dr Maureen McDonald, Dr Cathy Risdon, Dr Elkafi Hassini, Laura Cleghorn, and Dr Sonia Anand. We would also like to thank the Hamilton Family Health Team leadership for their guidance and continuous support of our work, Dr Brian McKenna and Gloria Jordan, as well as Elizaveta (Liza) Zvereva for her involvement in many of our projects. We would like to thank Dr Zahira Khalid, one of the founders of PEACH. Lastly, we would like to thank everyone at The Canadian Coalition for Green Health Care for their continued support of our work, specifically Linda Varangu, one of the founders of PEACH Health Ontario.

Contributors CQ, FP, IK, SK, RdS and MS all contributed equally to the paper conceptualisation and content editing. CQ, FP and IK led content preparation and writing. MS and SK designed the PEACH tree diagram. Authors contributed equally to this brief report. MS is the guarantor.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Not applicable.

Ethics approval Not applicable.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data sharing not applicable as no datasets generated and/or analysed for this study.

Open access This is an open access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: <http://creativecommons.org/licenses/by-nc/4.0/>.

ORCID iDs

Fiona Parascandalo <http://orcid.org/0000-0002-4013-6338>

Iliya Khakban <http://orcid.org/0009-0001-4524-7838>

REFERENCES

- 1 Braithwaite J, Smith CL, Leask E, *et al*. Strategies and tactics to reduce the impact of healthcare on climate change: systematic review. *BMJ* 2024;387:e081284.
- 2 Sergeant M, Webster R, Varangu L, *et al*. Identifying Opportunities for Greenhouse Gas Reductions and Cost Savings in Hospitals: A Knowledge Translation Tree. *Hcq* 2022;25:18–24.
- 3 Kates N, Ko E, Mathur N. Environmental stewardship and psychiatry—a plan for sustainable change. PEACH Health Ontario; 2024. Available: https://www.peachhealthontario.com/_files/ugd/f36758_12b92420ec4b4ec7b958ca14_1be957be.pdf
- 4 Arya N, Zigby J, Mah JJ, *et al*. Green Office Toolkit. PEACH Health Ontario, Sante Environment, CAPE, Pegasus Institute, McMaster University Department of Family Medicine. 2023. Available: https://www.peachhealthontario.com/_files/ugd/f36758_2e3a34a5d5bf46fdb4cc975f0ad9205b.pdf
- 5 Parascandalo F, Qian C, Khakban I, *et al*. Sustainable healthcare food system: business case and implementation guide. PEACH Health Ontario; 2024. Available: <https://peach.healthsci.mcmaster.ca/wp-content/uploads/2024/10/Sustainable-Health-Care-Food-System-Business-Case-Implementation-Guide-1.pdf>